

Team 10 Week 3 – Deliverable

Data Visualization Associate Internship

Problem Statement

Organizations offering multiple learning and career opportunities often lack a clear, unified view of opportunity performance and learner engagement across their Opportunity and Learner datasets. This project analyzes these datasets to evaluate opportunity performance, learner engagement, application trends, and reward/scholarship distribution, enabling data-driven decision-making through a validated, interactive dashboard.

1. KPI Selection & Rationale

The following eight KPIs were selected to directly address the areas outlined in the Week 3 brief: opportunity sign-up performance, completion rates, reward distribution, application trends over time, scholarship distribution, and lead-source performance. Each KPI is mapped below to the underlying field(s) it draws from, and the reasoning for selecting it.

KPI	Source Field(s)	Why It Was Selected
Sign-ups by opportunity (highest/lowest)	learner_full (joined), COUNT per opportunity	Directly identifies which opportunities are attracting the most/least learner interest — a core input for investment decisions.
Completion rate by opportunity	status_label = 'Rewards' as % of total	Sign-ups alone don't show whether learners follow through; completion rate reveals which opportunities retain learners to the end.
Reward count by opportunity	opportunity.reward_count	Shows how incentive structure varies across opportunities, and allows checking whether reward count correlates with sign-ups or completion.
Application counts, last 7 days	learner_opportunity.apply_date	Gives a short-term operational view of current application momentum.
Application trend, last 6 months	learner_opportunity.apply_date	Reveals medium-term seasonality or growth/decline patterns not visible in a single week.
Scholarship distribution by category	opportunity.microscholarship, category	Shows how financial support is allocated across opportunity types, relevant to

KPI	Source Field(s)	Why It Was Selected
		equity and budget planning.
Distribution by lead source	learner_opportunity.from_where_did_you_hear_about_us	Identifies which acquisition channels are driving the most applications, informing where to focus outreach.
Applicants count 2025 vs signup by opportunity 2026 % change	learner_opportunity.apply_date	Directly requested comparison; gives a concrete, quantified view of quarter-over-quarter momentum.

Note on Metric Definitions

- "Completion" is defined using the status_label = 'Rewards' outcome, since the dataset's literal 'Complete' status is used on only 4 records — too sparse to be a usable metric on its own. 'Rewards' represents an application that reached a completed, rewarded state, and is the closest reliable proxy in the data. This assumption should be confirmed with the data/system owner before being presented as a final business definition.
- Rankings by sign-up count and reward count are shown without a minimum sample-size filter for KPI 1 and 3 (to answer the literal 'highest and lowest' brief), but a minimum of 5 applications was applied to the completion-rate ranking (KPI 2), since completion rate on very small sample sizes (e.g. 1 out of 1) is not a statistically meaningful comparison.

2. Data Preparation Note

Before computing the KPIs below, one additional cleaning step was required beyond what was completed in Week 2: the from_where_did_you_hear_about_us field (needed for the lead-source KPI) contained five sets of inconsistent spelling/formatting for the same underlying source. These were standardized:

```
UPDATE learner_opportunity SET from_where_did_you_hear_about_us = 'Social Media'
WHERE from_where_did_you_hear_about_us = 'social Media';

UPDATE learner_opportunity SET from_where_did_you_hear_about_us = 'Illinois Institute of Technology'
WHERE from_where_did_you_hear_about_us = 'Illinois Tech';

UPDATE learner_opportunity SET from_where_did_you_hear_about_us = 'My Friend Or Relative'
WHERE from_where_did_you_hear_about_us = 'My Friend/Relative';

UPDATE learner_opportunity SET from_where_did_you_hear_about_us = 'Other'
WHERE from_where_did_you_hear_about_us = 'Others';

UPDATE learner_opportunity SET from_where_did_you_hear_about_us = 'Saint Louis University'
WHERE from_where_did_you_hear_about_us = 'SLU';
```

Additionally, while computing the scholarship distribution KPI, two further outlier values were identified in the microscholarship field — 500,000,000 (opportunity "KOREY119") and 10,000,000 (opportunity "dewf") — consistent with the same type of test/placeholder data entry error found in Week 2's fee field. These two records were excluded from the scholarship KPI calculation to avoid skewing category totals; they were not removed from the underlying table, only excluded from this specific aggregation, since they do not affect other metrics.

This field-level data quality issue (present since Week 1/2 but only surfaced now because this is the first KPI to use it) is worth a documented note for future data intake — the same category of issue (inconsistent text, occasional

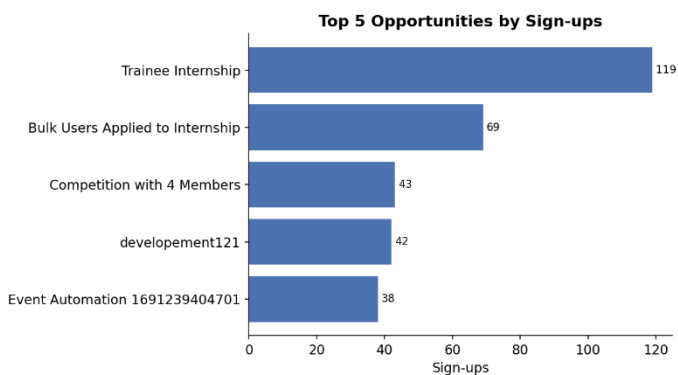
extreme numeric outliers) has now appeared in multiple unrelated fields, suggesting it is a systemic pattern in how data enters the source system rather than isolated incidents.

3. SQL Validation Queries & Results

Every metric intended for the dashboard was independently queried and validated here first, before being built as a visualization. This ensures that what appears on the dashboard has been checked against a known, reproducible SQL result.

3.1 KPI 1 — Sign-ups by Opportunity (Highest / Lowest)

```
SELECT o.name, COUNT(*) AS signups
FROM learner_opportunity lo
JOIN opportunity o
  ON REPLACE(lo.sk, 'Learner#', '') = REPLACE(o.opportunity_id, 'Opportunity#', '')
GROUP BY o.name
ORDER BY signups DESC
LIMIT 5;
```



Data Output

Messages

Notifications

SQL

	name text	signups bigint
1	Trainee Internship	119
2	Bulk Users Applied to Interns...	69
3	Competition with 4 Members	43
4	developement121	42
5	Event Automation 16912394...	38

Figure 1: Top 5 opportunities by sign-up count

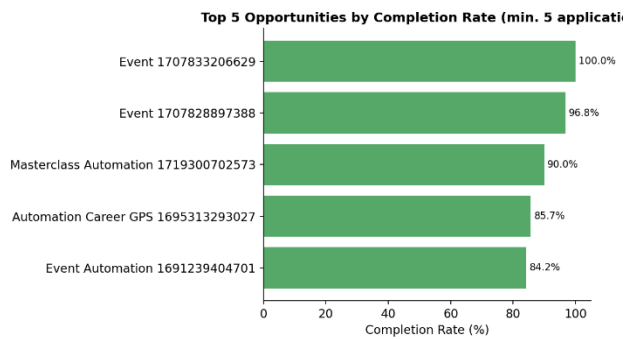
Rank	Opportunity	Sign-ups
1	Trainee Internship	119
2	Bulk Users Applied to Internship	69
3	Competition with 4 Members	43
4	developement121	42
5	Event Automation 1691239404701	38

At the lowest end, several opportunities are tied at exactly 1 sign-up each (e.g. AI Powered Data Insights, ttt, testing_fk, testingus24) — largely test or placeholder opportunity records rather than genuine low performers, and should be filtered out before this ranking is presented to business stakeholders as a "lowest performing" list.

3.2 KPI 2 — Completion Rate by Opportunity (minimum 5 applications)

```
SELECT o.name,
  COUNT(*) AS total,
  COUNT(*) FILTER (WHERE sm.status='Rewards') AS completed,
  ROUND(COUNT(*) FILTER (WHERE sm.status='Rewards')::numeric
    / COUNT(*) * 100, 1) AS completion_rate_pct
FROM learner_opportunity lo
JOIN opportunity o
  ON REPLACE(lo.sk, 'Learner#', '') = REPLACE(o.opportunity_id, 'Opportunity#', '')
JOIN status_mapping sm ON lo.status = sm.status_code
GROUP BY o.name
```

```
HAVING COUNT(*) >= 5
ORDER BY completion_rate_pct DESC
```



Data Output

Messages

Notifications

+

Showing rows: 1 to 524

Page No: 1

	name text	total bigint	completed bigint	completion_rate_pct numeric
1	Event 1707833206629	31	31	100.0
2	Event 1707828897388	30	30	100.0
3	Masterclass Automation 1719300702573	9	9	100.0
4	Test Moodle v2	5	5	100.0
5	Automation Career GPS 1695313293027	6	6	100.0
6	Xperience Design Hackathon: Shaping Memorable Moments	9	8	88.9
7	Event Automation 1691239404701	37	32	86.5
8	Career Automation 1687007545155	5	4	80.0
Total rows: 524 Query complete 00:00:00.304				

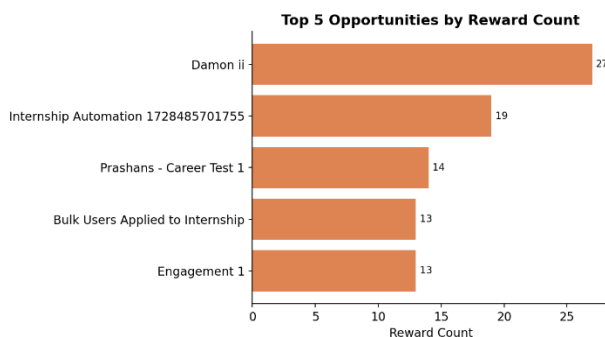
Figure 2: Top 5 opportunities by completion rate

Opportunity	Total Applications	Completed	Completion Rate
Event 1707833206629	31	31	100.0%
Event 1707828897388	30	30	100.0%
Masterclass Automation 1719300702573	9	9	100.0%
Automation Career GPS 1695313293027	6	6	100.0%
Event Automation 1691239404701	37	32	86.5%

At the lowest end (still with a minimum of 5 applications), several Masterclass opportunities show a 0% completion rate — none of their 5-10 applicants reached the 'Rewards' status.

3.3 KPI 3 — Reward Count by Opportunity

```
SELECT name, reward_count
FROM opportunity
ORDER BY reward_count DESC
LIMIT 5;
```



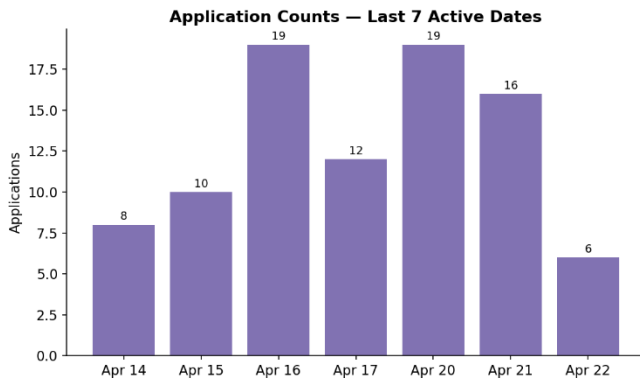
Data Output			Messages	Notifications
Showing rows: 1 to 524				Page No: 1
	name text	reward_count integer		
1	Damon ii	27		
2	Internship Automation 17284857017...	19		
3	Prashans - Career Test 1	14		
4	Cybersecurity: Defensive Hack	13		
5	Bulk Users Applied to Internship	13		

Figure 3: Top 5 opportunities by reward count

Highest: Damon ii (27 rewards), Internship Automation 1728485701755 (19), Prashans - Career Test 1 (14), Bulk Users Applied to Internship (13), Engagement 1 (13). Notably, 366 of the 5,713 opportunities (6.4%) have zero rewards attached at all.

3.4 KPI 4 — Application Counts, Last 7 Active Dates

```
SELECT apply_date::date AS day, COUNT(*)
FROM learner_opportunity
WHERE apply_date IS NOT NULL
GROUP BY day
ORDER BY day DESC
LIMIT 7;
```



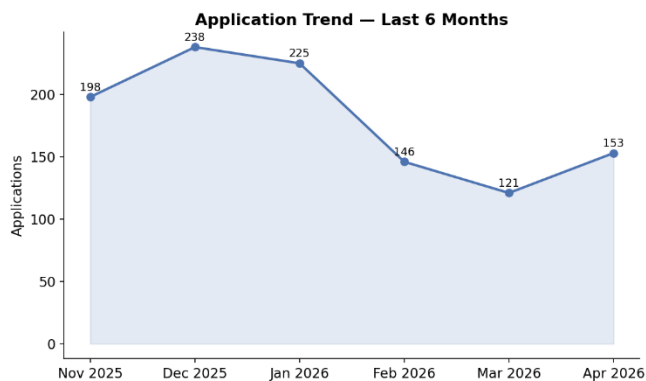
Data Output			Messages	Notifica
	day date	count bigint		
1	2026-04-22	6		
2	2026-04-21	16		
3	2026-04-20	19		
4	2026-04-17	12		
5	2026-04-16	19		
6	2026-04-15	10		
7	2026-04-14	8		

Figure 4: Application counts across the 7 most recent active dates in the dataset

Date	Applications
Apr 14, 2026	8
Apr 15, 2026	10
Apr 16, 2026	19
Apr 17, 2026	12
Apr 20, 2026	19
Apr 21, 2026	16
Apr 22, 2026	6

3.5 KPI 5 — Application Trend, Last 6 Months

```
SELECT DATE_TRUNC('month', apply_date) AS month, COUNT(*)
FROM learner_opportunity
WHERE apply_date IS NOT NULL
GROUP BY month
ORDER BY month DESC
LIMIT 6;
```



Data Output			Messages	Notifications
	month timestamp without time zone	count bigint		
1	2026-04-01 00:00:00	153		
2	2026-03-01 00:00:00	121		
3	2026-02-01 00:00:00	146		
4	2026-01-01 00:00:00	225		
5	2025-12-01 00:00:00	238		
6	2025-11-01 00:00:00	198		

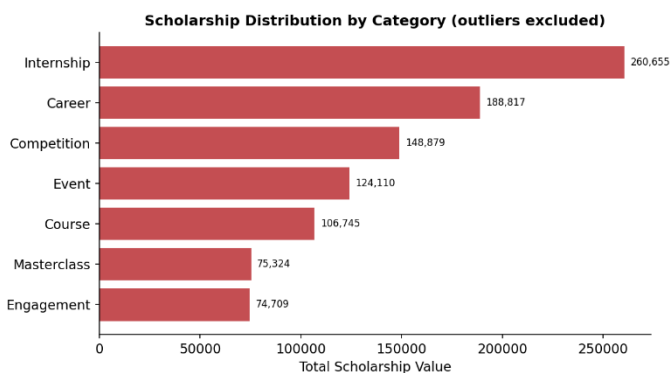
Figure 5: Application volume by month, most recent 6 months in the dataset

Month	Applications
Nov 2025	198
Dec 2025	238
Jan 2026	225
Feb 2026	146
Mar 2026	121
Apr 2026*	153

*April 2026 is a partial month as of the data extract date and should not be compared directly to full-month totals.

3.6 KPI 6 — Scholarship Distribution by Category

```
SELECT category, COUNT(*), SUM(microscholarship), ROUND(AVG(microscholarship),2)
FROM opportunity
WHERE microscholarship > 0 AND microscholarship <= 1000000
GROUP BY category
ORDER BY SUM(microscholarship) DESC;
```

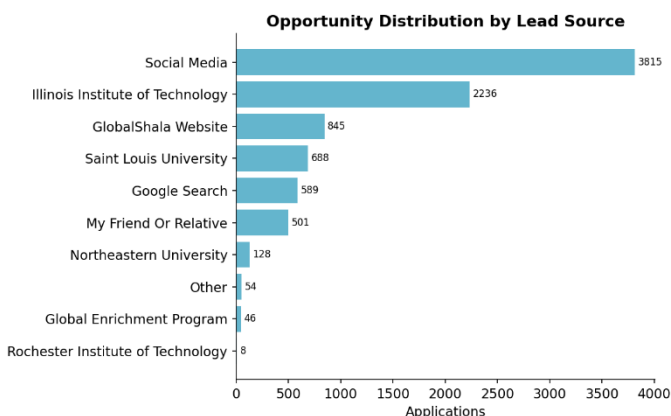


Data Output				
	category	count	sum	round
	text	bigint	bigint	numeric
1	Internship	1296	260655	201.12
2	Career	871	188817	216.78
3	Competition	666	148879	223.54
4	Event	549	124110	226.07
5	Course	558	106745	191.30
6	Masterclass	379	75324	198.74
7	Engagement	412	74709	181.33
8	JobSimulati...	167	20762	124.32
9	Xploreu	2	51	25.50
10	Program	2	49	24.50

Figure 6: Total scholarship value by opportunity category (outliers excluded)

3.7 KPI 7 — Opportunity Distribution by Lead Source

```
SELECT from_where_did_you_hear_about_us, COUNT(*)
FROM learner_opportunity
GROUP BY from_where_did_you_hear_about_us
ORDER BY COUNT(*) DESC;
```



Data Output		
	from_where_did_you_hear_about_us	count
	text	bigint
1	[null]	5868
2	Social Media	3815
3	Illinois Institute of Technology	2236
4	GlobalShala Website	845
5	Saint Louis University	688
6	Google Search	589
7	My Friend Or Relative	501
8	Northeastern University	128
9	Other	54
10	Global Enrichment Program	46
11	Rochester Institute of Technology	8
12	Hiring Platforms	1

Figure 7: Applications by lead source (standardized)

This field is populated on 60.3% of learner records; the remaining 39.7% have no recorded lead source and are excluded from this breakdown. This coverage gap should be noted alongside any lead-source chart on the dashboard.

3.8 KPI 8 — Applicants count 2025 vs signup by opportunity 2026 % change

```
SELECT
  COUNT(*) FILTER (WHERE apply_date >= '2025-10-01' AND apply_date < '2026-01-01') AS q4_2025,
  COUNT(*) FILTER (WHERE apply_date >= '2026-01-01' AND apply_date < '2026-04-01') AS q1_2026
FROM learner_opportunity;
```

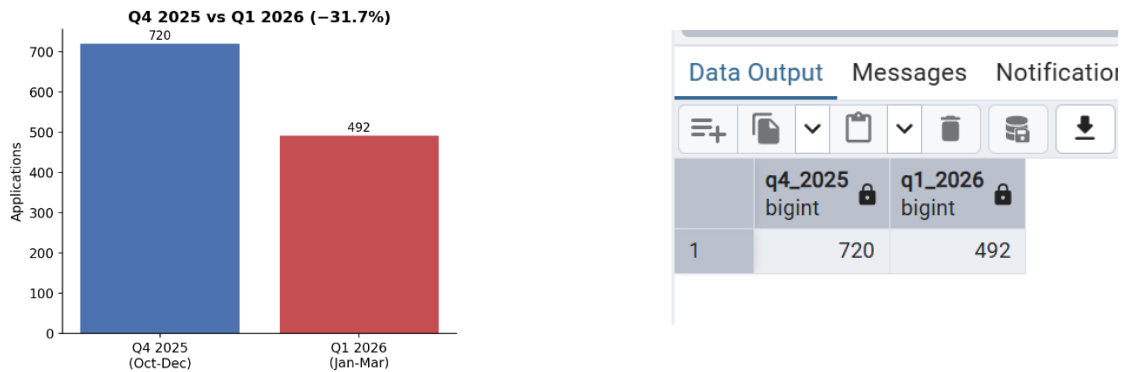


Figure 8: Application volume, Q4 2025 vs Q1 2026

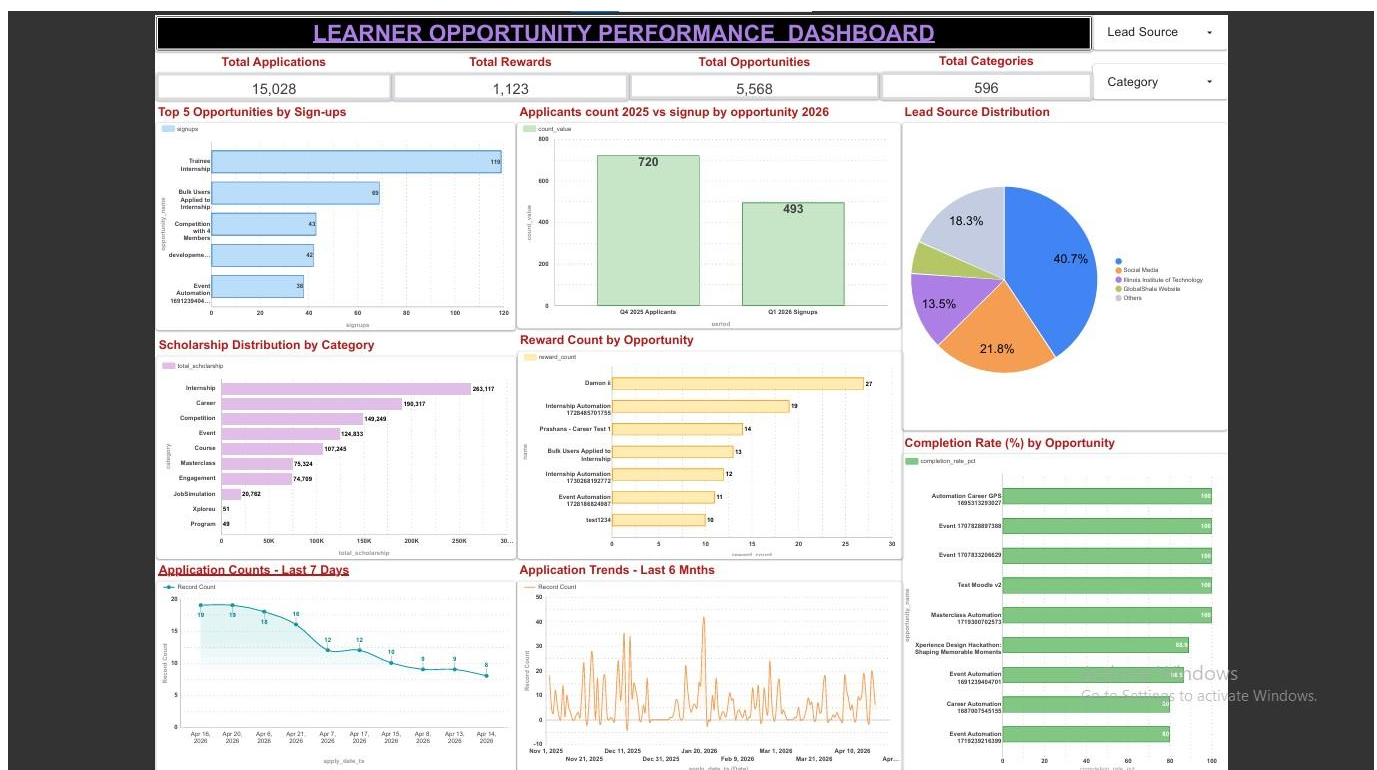
Application count 2025 (Oct-Dec): 720 applications. Signups by opportunity 2026 (Jan-Mar): 492 applications. This is a decline of 31.7% quarter-over-quarter.

4. Dashboard Accessibility & Design

4.1 Dashboard Accessibility

The Learner Opportunity Performance Dashboard was developed in Google Looker Studio and published as an interactive dashboard. It enables users to explore learner applications, opportunity performance, rewards, scholarship distribution, completion rates, and lead source analysis through a single interface.

Dashboard Link: <https://datastudio.google.com/reporting/a68d096d-9ee9-4915-bf72-06932f45aeb4>



Accessibility Features

- Interactive global filters (Lead Source and Category)
- Hover tooltips for all charts
- KPI summary cards for quick insights
- Responsive layout with consistent colour scheme
- Charts refresh automatically when filters are applied

4.2 Dashboard Design Overview

Dashboard Objective

The dashboard provides a centralized view of learner engagement and opportunity performance by combining applications, rewards, scholarships, completion rates, lead sources, and time-based trends into one interactive analytical report. The design helps stakeholders monitor platform performance and make informed business decisions.

KPIs Included

- Total Applications – 15,028
- Total Rewards – 1,123
- Total Opportunities – 5,568
- Total Categories – 596

Dashboard Layout

Top Section

- KPI Cards

Middle Section

- Top 5 Opportunities by Sign-ups
- Applicants Count (Q4 2025 vs Q1 2026)
- Lead Source Distribution

Lower Section

- Scholarship Distribution by Category
- Reward Count by Opportunity
- Completion Rate by Opportunity

Bottom Section

- Application Counts (Last 7 Days)
- Application Trends (Last 6 Months)

Filters Used

- Lead Source
- Category

Reason for Dashboard Design

KPI cards provide an instant summary of overall platform activity. Horizontal bar charts compare opportunities and categories, pie charts show proportional distributions, while line charts present short-term and long-term trends. Interactive filters allow users to focus on specific business areas without changing the dashboard layout.

5. Key insights and Recommendations

Trend Analysis

Includes

- Application Counts (Last 7 Days)

- Application Trends (Last 6 Months)

These charts help monitor recent activity and long-term engagement.

Filters Used

Two global dashboard filters were implemented.

Lead Source

Allows users to analyse learner behaviour based on different acquisition channels.

Examples include

- Social Media
- Google Search
- GlobalShala Website
- Illinois Institute of Technology

Category

Allows filtering of opportunities by category such as

- Internship
- Competition
- Event
- Career
- Masterclass

This enables focused analysis without creating multiple dashboards.

Visualization	Purpose
KPI Cards	Summary statistics
Horizontal Bar Chart	Ranking opportunities
Vertical Bar Chart	Quarterly comparison
Pie Chart	Lead source distribution
Horizontal Bar Chart	Scholarship distribution
Horizontal Bar Chart	Reward count
Horizontal Bar Chart	Completion rate

Line Chart	Recent applications
Time Series Line Chart	Long-term application trend

Why These Visualizations Were Selected

Different chart types were selected according to the nature of the data.

Bar charts effectively compare opportunities and categories.

Pie charts clearly display proportional distributions.

Line charts are ideal for showing application trends over time.

KPI cards allow management to quickly understand the overall performance without exploring detailed charts.

Global filters make the dashboard interactive while maintaining a clean layout.

Findings and Recommendations

Key Findings

- **Overall Dashboard**

The platform contains 15,028 learner applications, 5,568 opportunities, and 1,123 rewards, indicating a large and active ecosystem of learning opportunities. The dashboard provides a consolidated view of learner engagement and opportunity performance.

- **Opportunity Performance**

The Trainee Internship opportunity recorded the highest number of sign-ups with 119 applications, followed by Bulk Users Applied to Internship with 69 applications. This suggests that internship-oriented opportunities attract the highest learner interest.

Several opportunities received fewer than five applications, indicating limited learner engagement. Many of these appear to be testing or inactive opportunities and may require review before being included in business reporting.

- **Reward Distribution**

The opportunity Damon ii recorded the highest reward count with 27 rewards, while several other opportunities achieved reward counts between 10 and 19.

A significant number of opportunities have relatively low reward allocations, indicating opportunities to improve learner incentives.

- **Scholarship Analysis**

Internship opportunities account for the highest scholarship allocation (approximately 263,117), followed by Career and Competition categories.

Categories such as Program and XploreU receive minimal scholarship funding, suggesting that financial support is concentrated in a limited number of opportunity types.

- **Completion Rate**

Several opportunities achieved a 100% completion rate, demonstrating excellent learner engagement once enrolled.

However, other opportunities remain below 80%, indicating that learners may disengage before reaching the reward stage.

Completion rate should be monitored alongside sign-up volume because high enrolment does not always translate into successful completion.

- **Lead Source Analysis**

Social Media contributes the largest share of learner acquisition (40.7%), followed by Social Media campaigns and Illinois Institute of Technology referrals.

This indicates that digital marketing remains the most effective recruitment channel.

- **Application Trends**

Recent application activity shows consistent learner participation with daily applications ranging between 6 and 19.

Over the last six months, application activity fluctuated considerably, with noticeable peaks during December and January.

The quarterly comparison also shows that applications declined from 720 in Q4 2025 to 493 in Q1 2026, representing a reduction of approximately 31.5%. This may indicate seasonal variation or reduced campaign effectiveness.

Recommendations

- **Improve Underperforming Opportunities**

Review opportunities receiving very low application counts to determine whether they require better promotion, updated content, or removal if they are no longer active.

- **Increase Completion Rates**

Investigate opportunities with completion rates below 80% to identify potential barriers such as lengthy application processes, insufficient learner support, or unclear reward criteria. Optimize Marketing Channels

Since Social Media contributes the highest proportion of applications, additional investment in digital campaigns may further improve learner acquisition. At the same time, lower-performing channels should be evaluated to improve return on marketing expenditure.

- **Strengthen High-Performing Categories**

Internship opportunities consistently perform well across sign-ups and scholarship allocation. Expanding similar opportunities may increase overall learner engagement.

- **Monitor Seasonal Trends**

The observed decline in Q1 2026 suggests that application activity should be monitored regularly. Future campaigns can be scheduled before historically high-engagement periods to maximize participation.

- **Improve Data Quality**

Several records contain test opportunities, inconsistent naming conventions, and missing lead source values. Strengthening data validation during data entry will improve dashboard accuracy and reporting reliability.